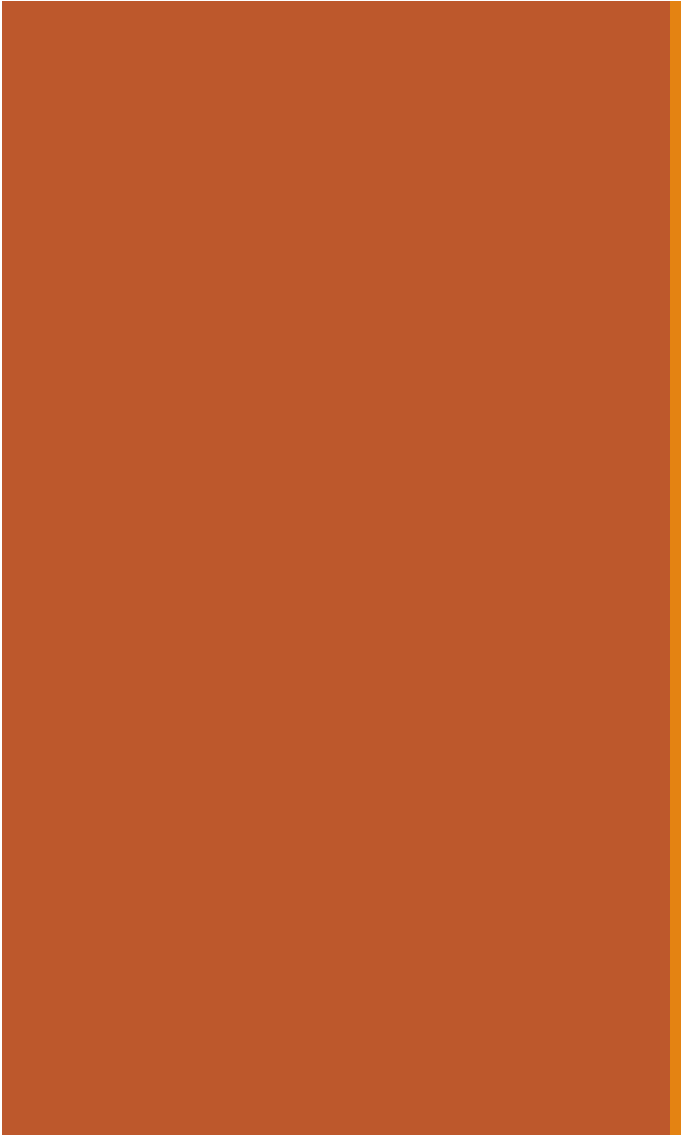


16: Continuous Joint Distributions

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May 6th, 2024

[Lecture Discussion on Ed](#)



Continuous Joint Distributions

Stanford logo with darts



The Stanford letterhead logo was created by throwing 500,000 darts according to a joint distribution.

not really, but let's pretend

If we throw another dart according to the same distribution, what is

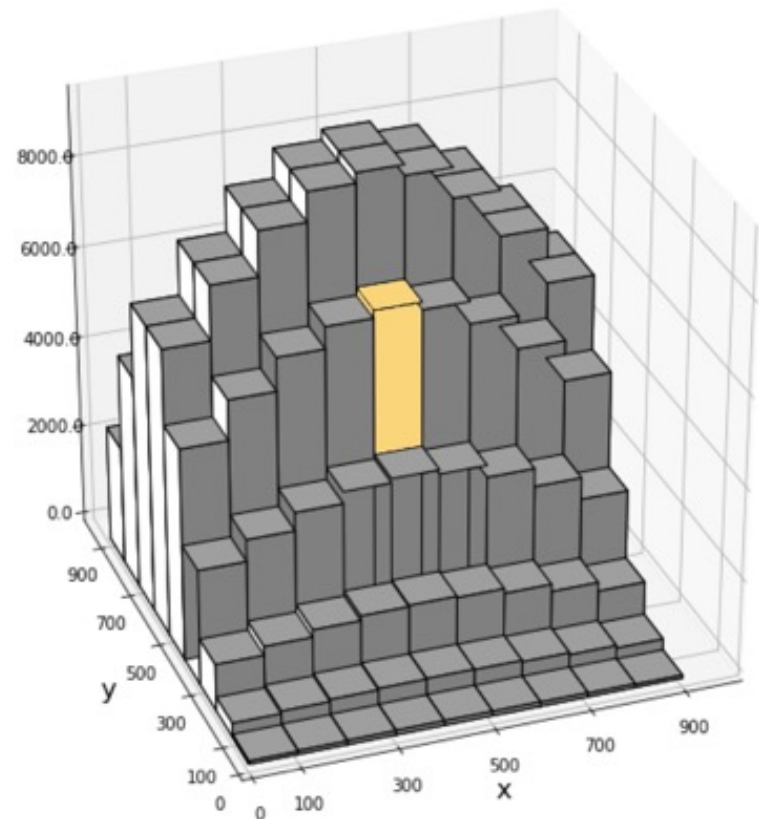
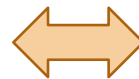
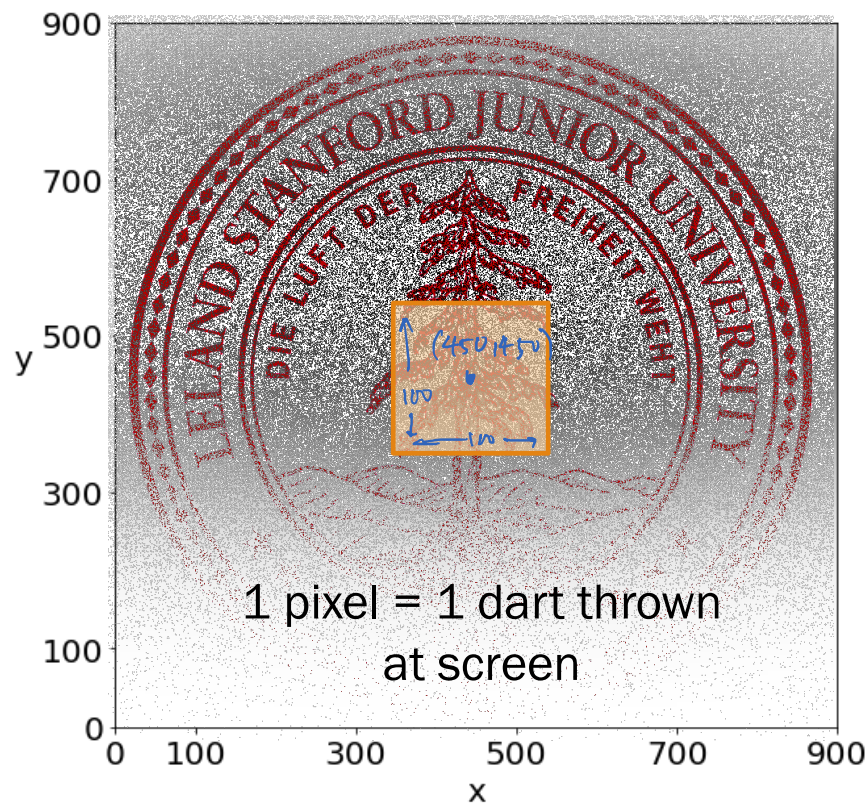
$P(\text{dart hits within } r \text{ pixels of center})?$

Quick check: What is the probability that a dart hits at (456.2344132343, 532.1865739012)?

zero

CS109 logo with darts

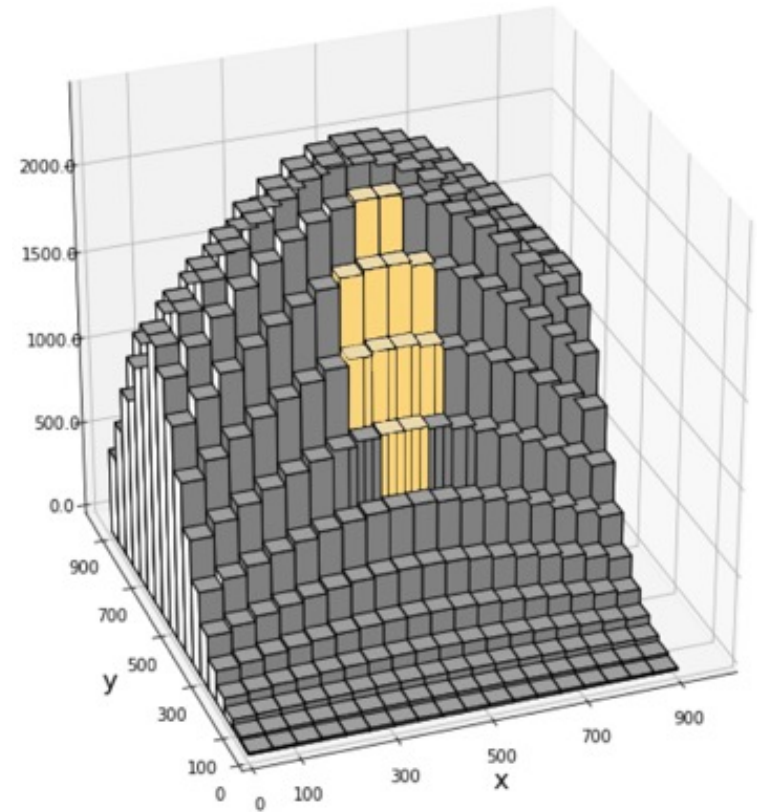
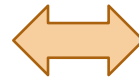
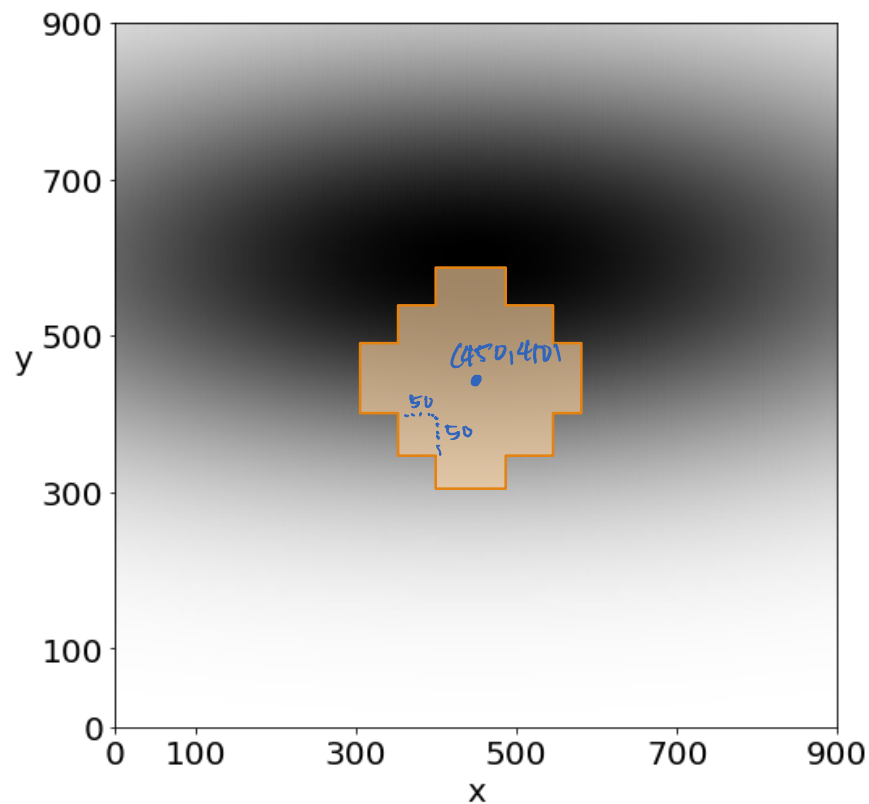
$P(\text{dart hits within } r \text{ pixels of center})?$



Possible dart counts (in 100x100 boxes)

CS109 logo with darts

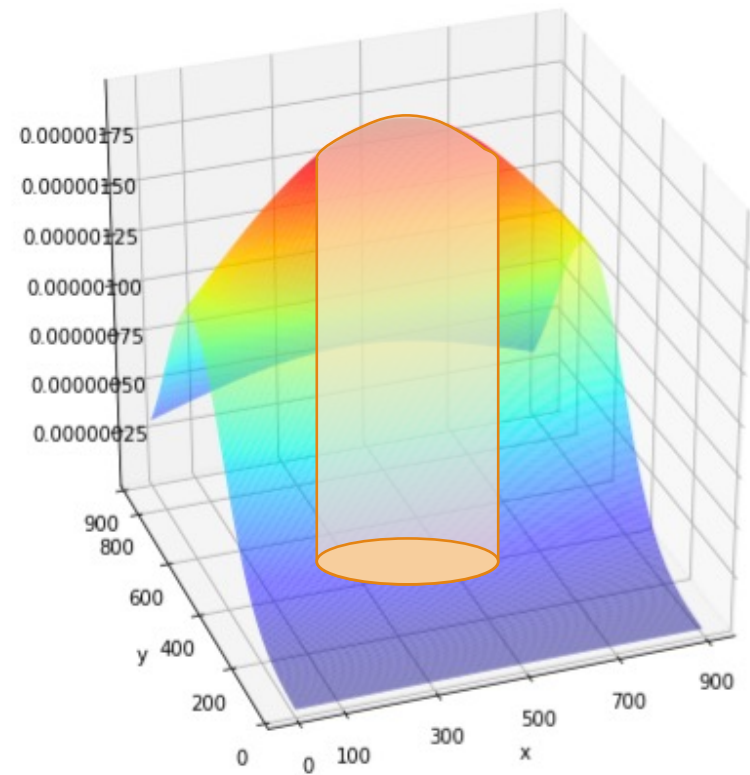
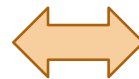
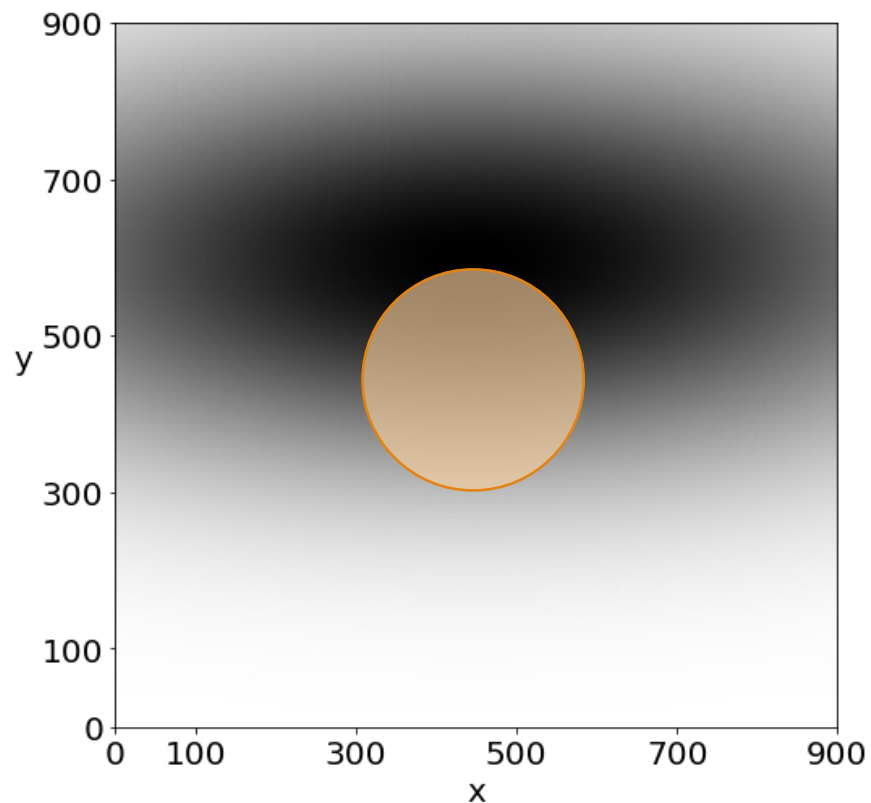
$P(\text{dart hits within } r \text{ pixels of center})?$



Possible dart counts (in 50x50 boxes)

CS109 logo with darts

$P(\text{dart hits within } r \text{ pixels of center})?$



Possible dart counts
(in infinitesimally small boxes) **iversity**

Continuous joint probability density functions

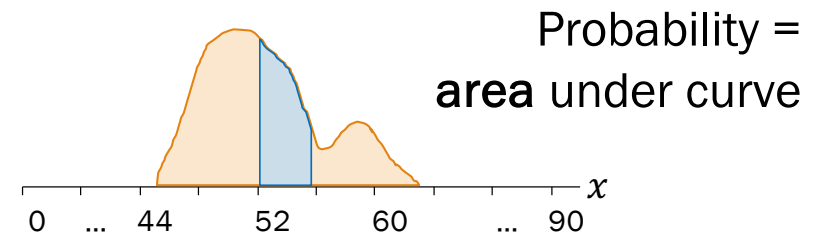
If two random variables X and Y are jointly continuous, then there exists a **joint probability density function** $f_{X,Y}$ defined over $-\infty < x, y < \infty$ such that:

$$\underbrace{P(a_1 \leq X \leq a_2, b_1 \leq Y \leq b_2)}_{\text{this is a probability}} = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \underbrace{f_{X,Y}(x, y)}_{\text{2D density, e.g. } \frac{\text{prob}}{\text{pixel}^2}} dy dx$$

From one continuous RV to jointly continuous RVs

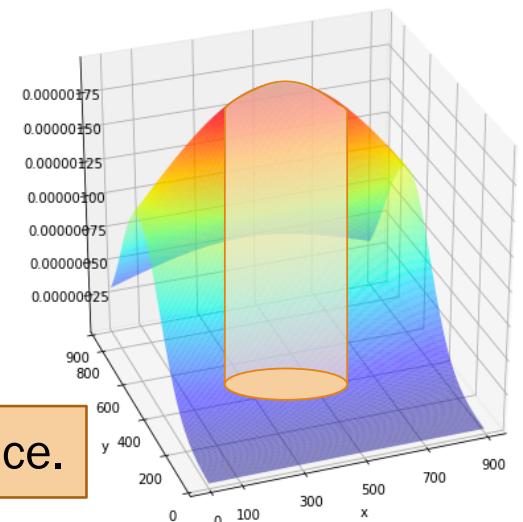
Single continuous RV X

- PDF f_X such that $\int_{-\infty}^{\infty} f_X(x) dx = 1$
- Integrate to get probabilities



Jointly continuous RVs X and Y

- PDF $f_{X,Y}$ such that $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy dx = 1$
- Double integrate to get probabilities



pair of
^
Probability for jointly continuous RVs is **volume** under a surface.

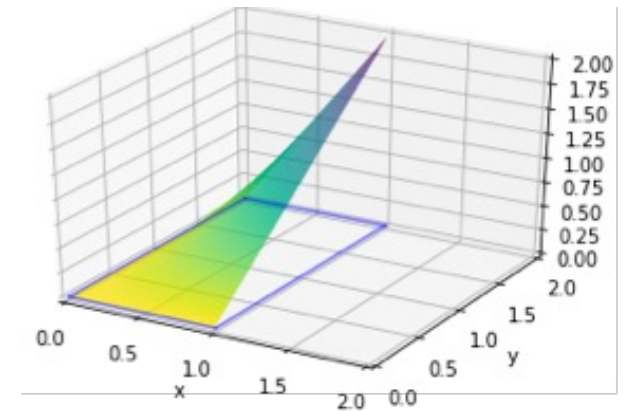
Double integrals without tears

Let X and Y be two continuous random variables.

- Support: $0 \leq X \leq 1$, $0 \leq Y \leq 2$.

Is $g(x, y) = xy$ a valid joint PDF over X and Y ?

Write down the definite double integral that must integrate to 1:



Stanford University

Double integrals without tears

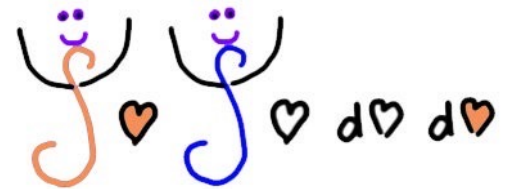
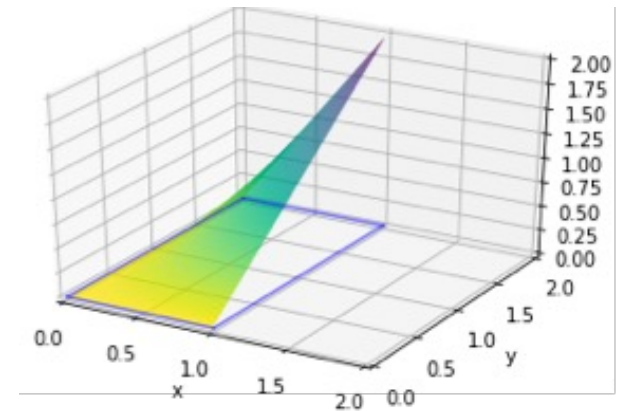
Let X and Y be two continuous random variables.

- Support: $0 \leq X \leq 1$, $0 \leq Y \leq 2$.

Is $g(x, y) = xy$ a valid joint PDF over X and Y ?

Write down the definite double integral that must integrate to 1:

$$\underbrace{\int_{y=0}^2 \int_{x=0}^1 xy \, dx \, dy = 1}_{\text{(used in next slide)}} \quad \text{or} \quad \int_{x=0}^1 \int_{y=0}^2 xy \, dy \, dx = 1$$



Double integrals without tears

Let X and Y be two continuous random variables.

- Support: $0 \leq X \leq 1, 0 \leq Y \leq 2$.

Is $g(x, y) = xy$ a valid joint PDF over X and Y ?

0. Set up integral:

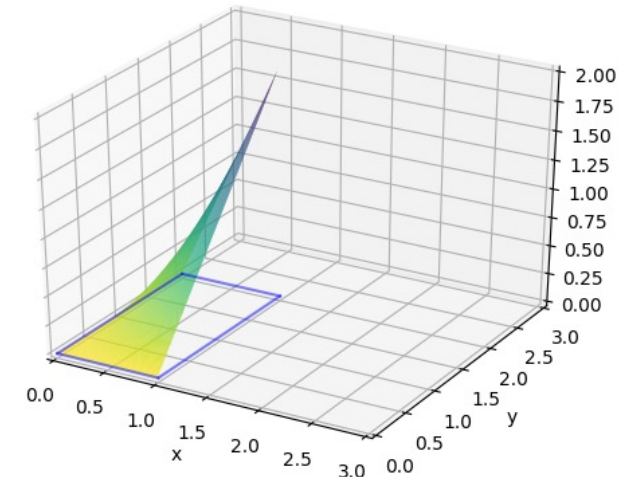
$$1 \stackrel{?}{=} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) dx dy = \int_{y=0}^2 \int_{x=0}^1 xy dx dy$$

1. Evaluate inside integral by treating y as a constant:

$$\int_{y=0}^2 \left(\int_{x=0}^1 xy dx \right) dy = \int_{y=0}^2 y \left(\int_{x=0}^1 x dx \right) dy = \int_{y=0}^2 y \left[\frac{x^2}{2} \right]_0^1 dy = \int_{y=0}^2 y \frac{1}{2} dy$$

2. Evaluate remaining (single) integral:

$$\int_{y=0}^2 y \frac{1}{2} dy = \left[\frac{y^2}{4} \right]_{y=0}^2 = 1 - 0 = 1$$

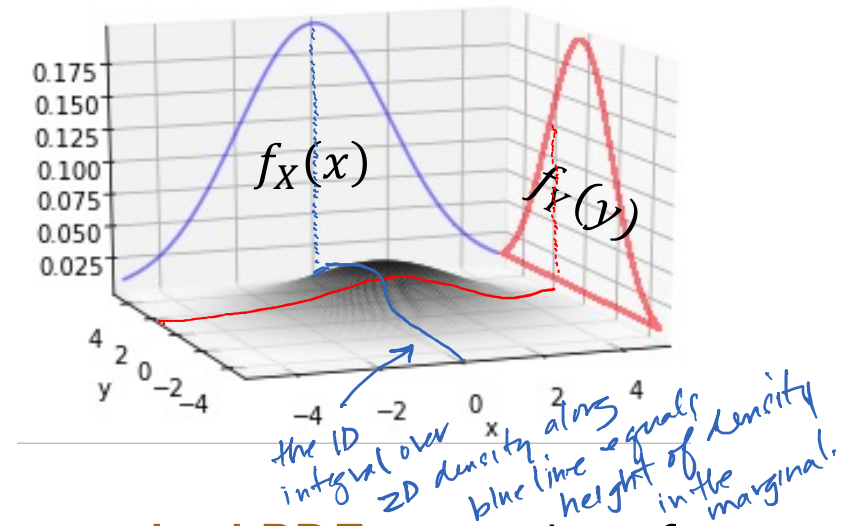


Yes, $g(x, y)$ is a valid joint PDF because it integrates to 1.

Marginal distributions

Suppose X and Y are continuous random variables with joint PDF:

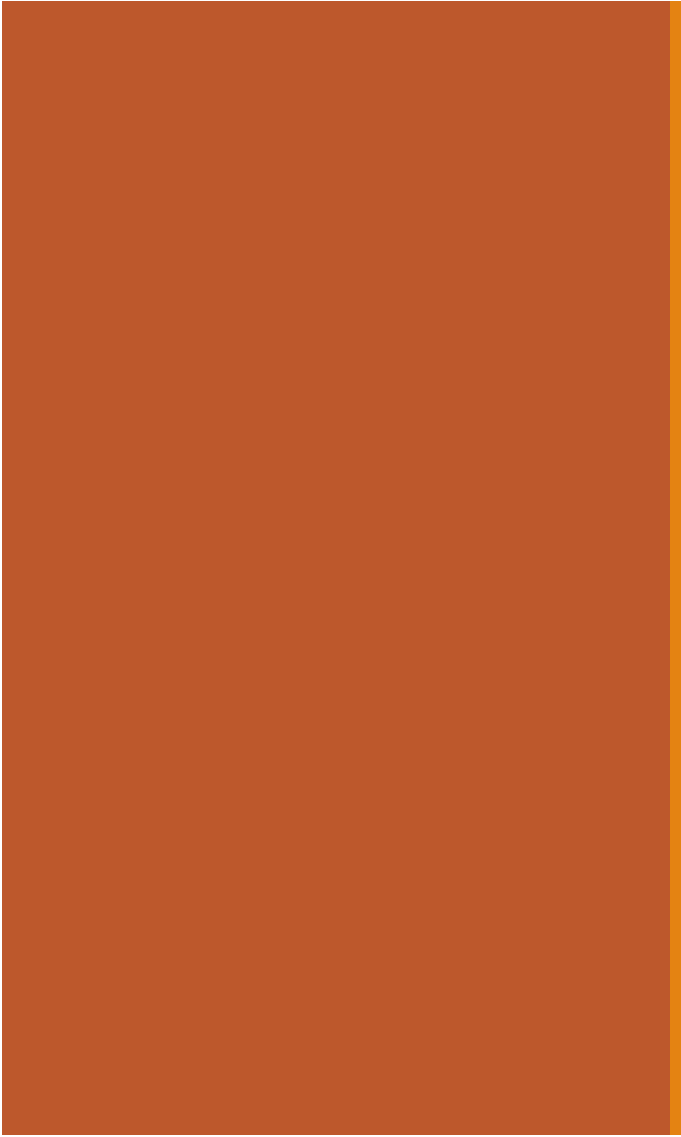
$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy dx = 1$$



The marginal density functions—that is, the **marginal PDFs**—are therefore:

$$f_X(a) = \int_{-\infty}^{\infty} f_{X,Y}(a, y) dy$$

$$f_Y(b) = \int_{-\infty}^{\infty} f_{X,Y}(x, b) dx$$



Joint CDFs

CDFs and PDFs in one dimension

Review

For a one-dimensional continuous random variable X with PDF f , the CDF (cumulative distribution function) is

$$F(a) = P(X \leq a) = \int_{-\infty}^a f(x) dx$$

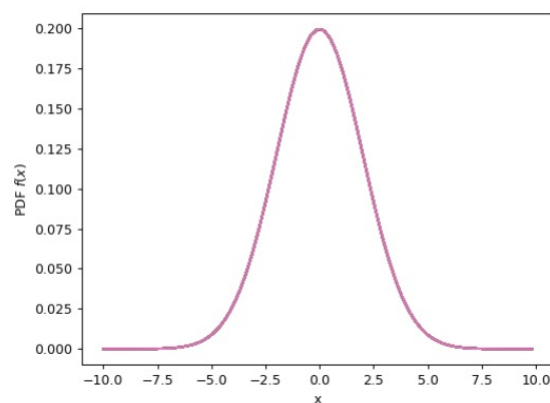
The density f is the derivative of the CDF, F :

$$f(x) = \frac{d}{dx} F(x)$$

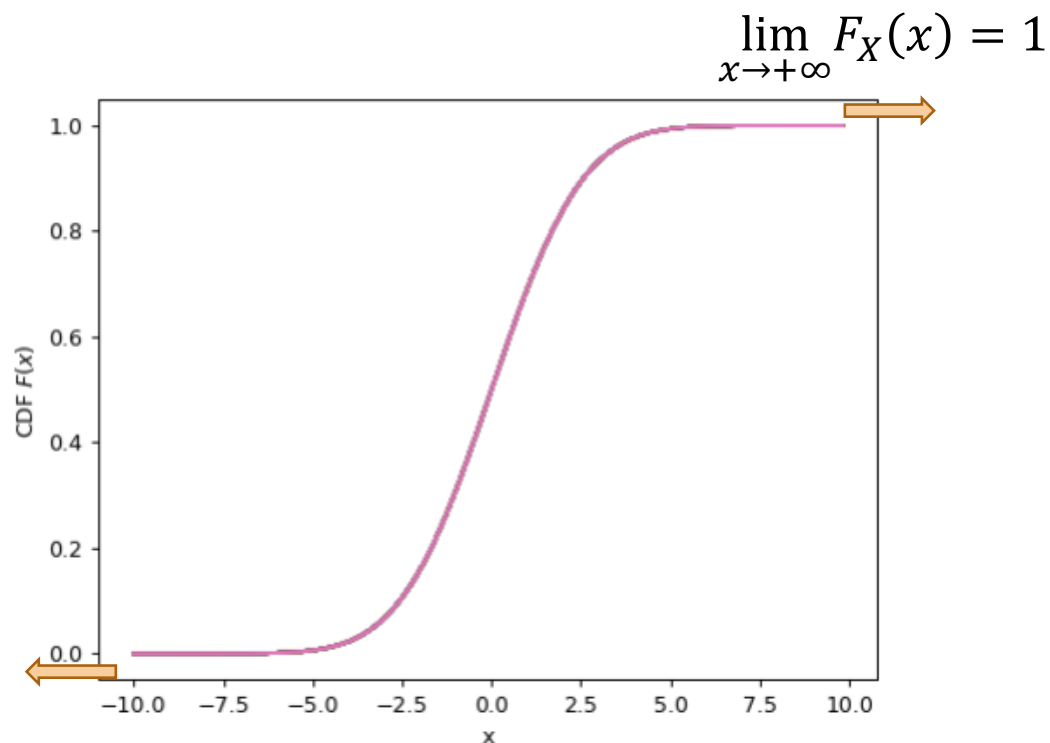
Fundamental Theorem
of Calculus

Single variable CDF, graphically

Review



$f_X(x)$



$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow +\infty} F_X(x) = 1$$

$$F_X(x) = P(X \leq x)$$

Joint cumulative distribution function

For two random variables X and Y , there can be a **joint cumulative distribution function** $F_{X,Y}$:

$$F_{X,Y}(a, b) = P(X \leq a, Y \leq b)$$

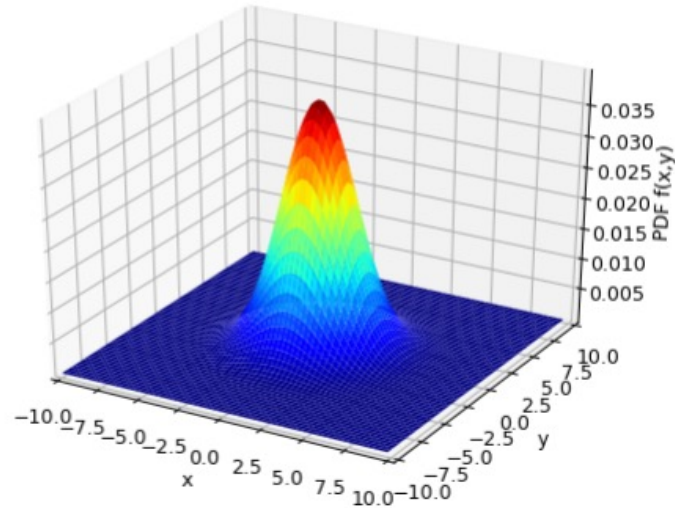
For discrete X and Y :

$$F_{X,Y}(a, b) = \sum_{x \leq a} \sum_{y \leq b} p_{X,Y}(x, y)$$

For continuous X and Y :

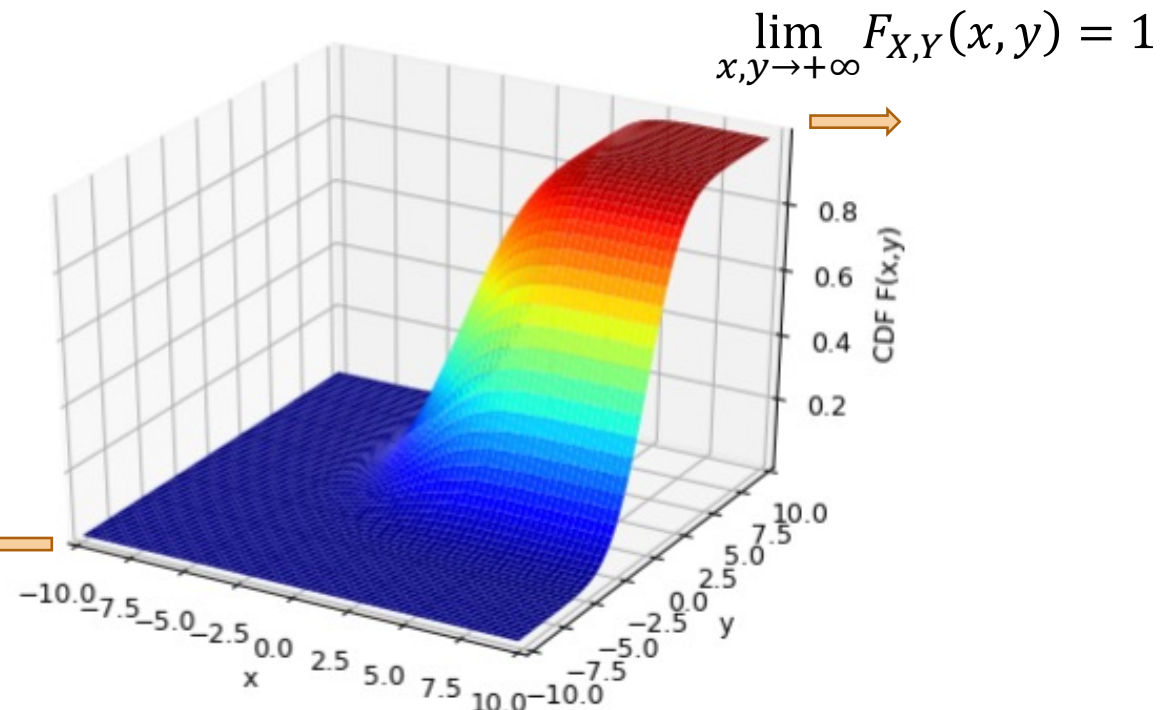
$$F_{X,Y}(a, b) = \int_{-\infty}^a \int_{-\infty}^b f_{X,Y}(x, y) dy dx$$
$$f_{X,Y}(a, b) = \frac{\partial^2}{\partial a \partial b} F_{X,Y}(a, b)$$

Joint CDF, graphically

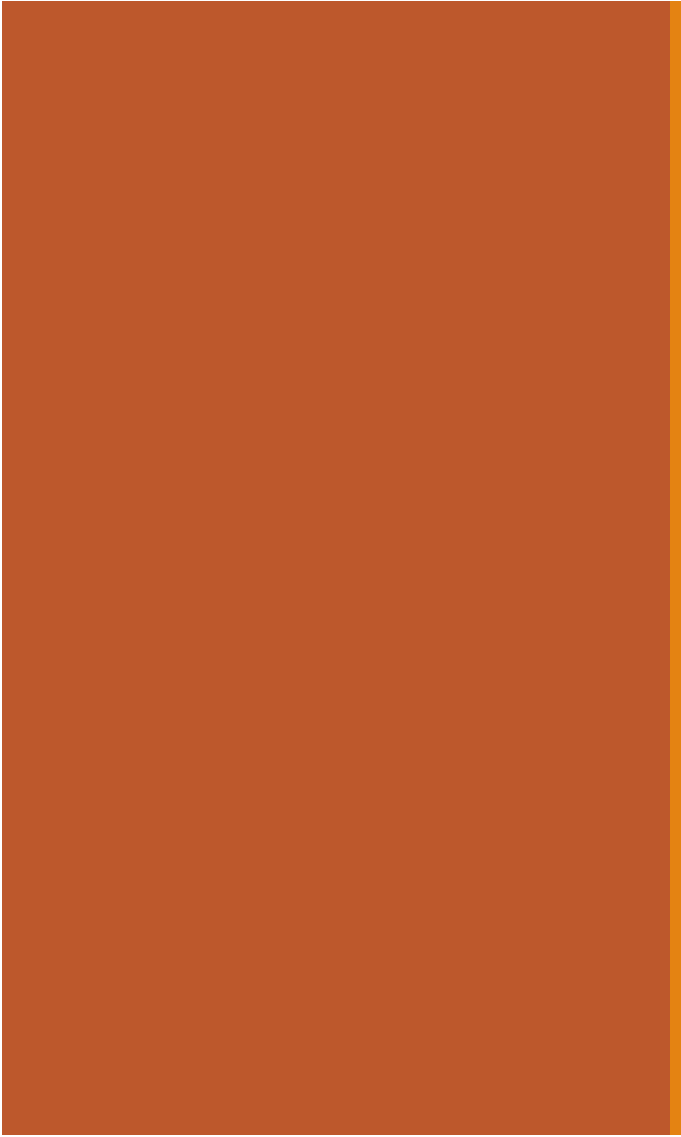


$$\lim_{x,y \rightarrow -\infty} F_{X,Y}(x,y) = 0$$

$$f_{X,Y}(x,y)$$



$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$



Independent Continuous RVs

Independent continuous RVs

Two continuous random variables X and Y are **independent** if:

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y) \quad \forall x, y$$

Equivalently:

$$\begin{aligned} F_{X,Y}(x, y) &= F_X(x)F_Y(y) \\ f_{X,Y}(x, y) &= f_X(x)f_Y(y) \end{aligned} \quad \forall x, y$$

Proof for PDF:

$$\begin{aligned} f_{X,Y}(x, y) &= \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_X(x)F_Y(y) \\ &= \frac{\partial}{\partial x} \frac{\partial}{\partial y} F_X(x)F_Y(y) = \frac{\partial}{\partial x} F_X(x) \frac{\partial}{\partial y} F_Y(y) \\ &= f_X(x)f_Y(y) \end{aligned}$$

Handwritten notes:
- Blue: "constant with respect to x" (under $F_X(x)$ in the second line)
- Green: "constant with respect to y" (under $F_Y(y)$ in the second line)

Independent continuous RVs

Two continuous random variables X and Y are **independent** if:

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y) \quad \forall x, y$$

Equivalently:

$$\begin{aligned} F_{X,Y}(x, y) &= F_X(x)F_Y(y) \\ f_{X,Y}(x, y) &= f_X(x)f_Y(y) \end{aligned} \quad \forall x, y$$

More generally, X and Y are **independent** if the joint PDF factors into two, single-variable marginal probability densities:

$$f_{X,Y}(x, y) = g(x)h(y), \text{ where } -\infty < x, y < \infty$$

states that x and y
can freely roam over
their full support without
constraint.

Pop quiz! (just kidding)

$f_{X,Y}(x,y) = g(x)h(y)$,
where $-\infty < x, y < \infty$  independent
 X and Y

Are X and Y independent in the following cases?

1. $f_{X,Y}(x,y) = 6e^{-3x}e^{-2y}$
where $0 < x, y < \infty$

2. $f_{X,Y}(x,y) = 4xy$
where $0 < x, y < 1$

3. $f_{X,Y}(x,y) = 24xy$
where $0 < x + y < 1$



Pop quiz! (just kidding)

$f_{X,Y}(x,y) = g(x)h(y)$,
where $-\infty < x, y < \infty$  independent
X and Y

Are X and Y independent in the following cases?

✓ 1. $f_{X,Y}(x,y) = 6e^{-3x}e^{-2y}$
where $0 < x, y < \infty$

Separable functions: $g(x) = 3e^{-3x}$
 $h(y) = 2e^{-2y}$

*both marginals
are exponentials*

✓ 2. $f_{X,Y}(x,y) = 4xy$
where $0 < x, y < 1$

Separable functions: $g(x) = 2x$
 $h(y) = 2y$

$\int_0^1 \int_0^1 4xy \, dx \, dy = 1 \Rightarrow$

$\int_0^1 C 4x \, dx \int_0^1 \frac{1}{C} 4y \, dy$

*C must be
chosen s.t.
that
yellow
and green
each
integrate
to 1.*

✗ 3. $f_{X,Y}(x,y) = 24xy$
where $0 < x + y < 1$

Cannot capture constraint on $x + y$!

*if separate
joint density
into two marginals.*

If you can factor densities over the entire support, you have independence.

More pop quiz! (more kidding)

X and Y have the following joint PDF:

$$f_{X,Y}(x,y) = 3e^{-3x}$$

where $0 < x < \infty, 1 < y < 2$

1. Are X and Y independent?
2. What is the marginal PDF of X ? Of Y ?
3. What is $E[X + Y]$?



More pop quiz! (more kidding)

X and Y have the following joint PDF:

$$f_{X,Y}(x,y) = 3e^{-3x}$$

where $0 < x < \infty, 1 < y < 2$

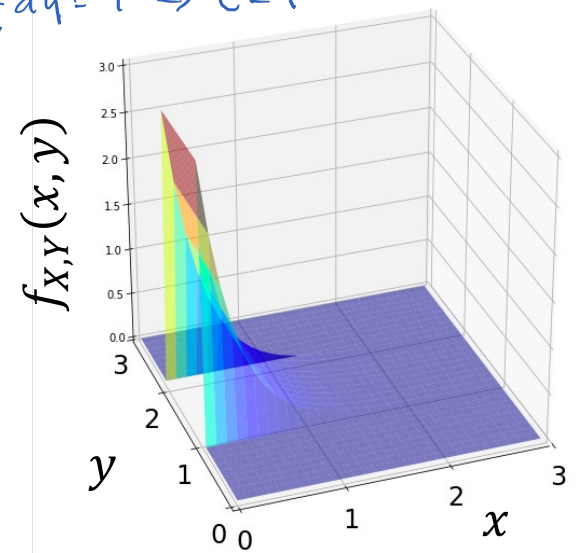
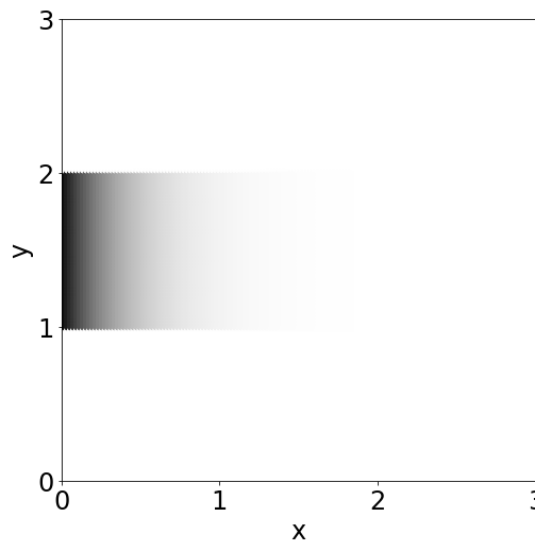
1. Are X and Y independent? 

$$g(x) = 3Ce^{-3x}, 0 < x < \infty \quad C \text{ is a constant}$$
$$h(y) = 1/C, \quad 1 < y < 2$$

$$\int_1^2 \frac{1}{C} dy = 1 \Rightarrow C = 1$$

2. What is the marginal PDF of X ? Of Y ?

3. What is $E[X + Y]$?



More pop quiz! (more kidding)

X and Y have the following joint PDF:

$$f_{X,Y}(x,y) = 3e^{-3x}$$

where $0 < x < \infty, 1 < y < 2$

1. Are X and Y independent? 

$$g(x) = 3e^{-3x}, 0 < x < \infty$$

$$h(y) = 1, 1 < y < 2$$

marginal PDF of X is $g(x) = 3e^{-3x}$, $X \sim \text{Exp}(3)$
marginal PDF of Y is $h(y) = 1$, $Y \sim \text{Uni}(1,2)$

2. What is the marginal PDF of X ? Of Y ?

strategy 1: $E[X+Y] = \int_1^2 \int_0^\infty (x+y) 3e^{-3x} dx dy$
by LOTUS

note: this annotation updated 5/20/24

3. What is $E[X + Y]$?

strategy 2: $E[X+Y] = E[X] + E[Y]$

$$= \frac{1}{3} + \frac{3}{2} = \frac{11}{6}$$

by linearity of expectation

The joy of meetings

Two people set up a meeting time. Each arrives independently at a time uniformly distributed between 12pm and 12:30pm.

Define $X = \#$ minutes past 12pm that person 1 arrives. $X \sim \text{Uni}(0, 30)$ $g(x) = \frac{1}{30}$
 $Y = \#$ minutes past 12pm that person 2 arrives. $Y \sim \text{Uni}(0, 30)$ $h(y) = \frac{1}{30}$

What is the probability that the first to arrive waits >10 mins for the other?

Compute: $P(X + 10 < Y) + P(Y + 10 < X) = 2P(X + 10 < Y)$ (by symmetry)

1. What is symmetry here?
2. How do we integrate to compute this probability?

$$f(x, y) = \left(\frac{1}{30}\right)^2$$
$$P = \iint_{\substack{0 \leq x, y \leq 30 \\ x + 10 < y}} \left(\frac{1}{30}\right)^2 dx dy + \iint_{\substack{0 \leq x, y \leq 30 \\ y + 10 < x}} \left(\frac{1}{30}\right)^2 dx dy$$

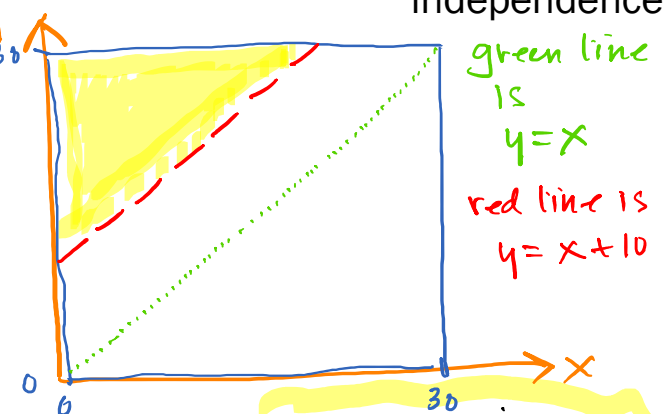
Double integrals: A guide

From last slide: $2P(X + 10 < Y) = 2 \iint_{\substack{x+10 < y, \\ 0 \leq x, y, \leq 30}} (1/30)^2 dx dy$ (by symmetry, independence)

Steps:

1. Draw a picture.
2. Set bounds "from outside in".
 - Outer integral bounds should be full range possible
 - Inner integral can depend on integration variable of outer integral

blue square
delineates full
range of PDF
in two dimensions

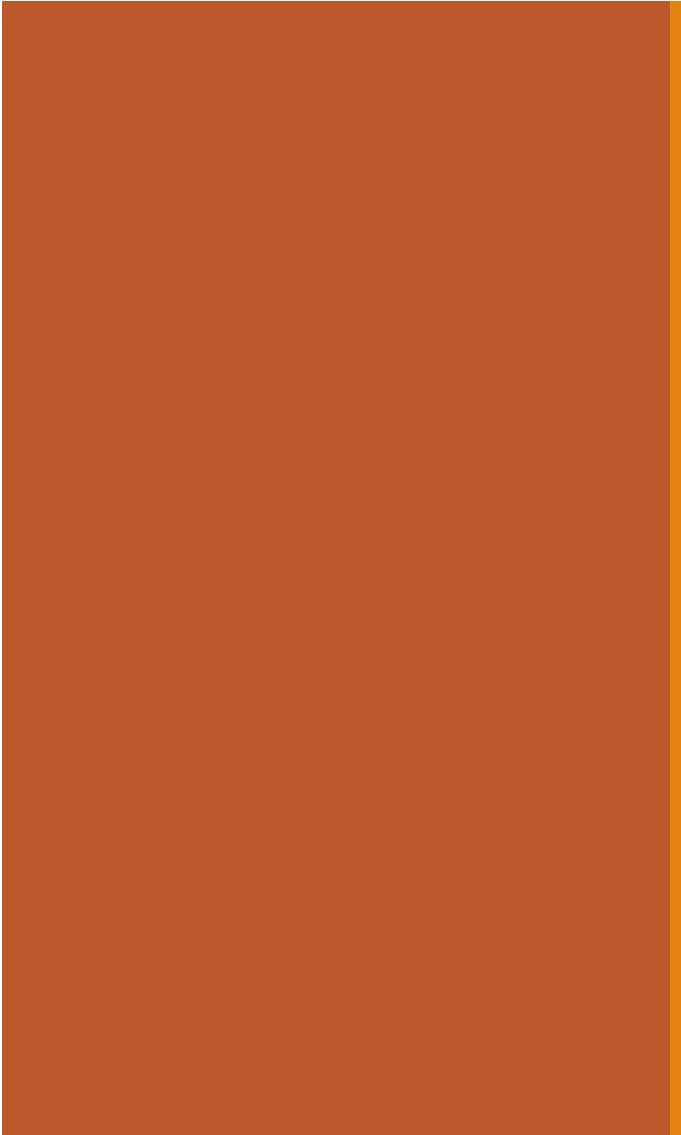


$$= \frac{2}{30^2} \int_{10}^{30} \int_0^{y-10} dx dy$$

$$= \frac{2}{30^2} \int_{10}^{30} (y-10) dy$$

$$= \dots = \frac{4}{9}$$

yellow region
identifies all points
where $y > x+10$



Sum of Independent Gaussians

Sum of independent Gaussians

$$\begin{aligned} X &\sim \mathcal{N}(\mu_1, \sigma_1^2), \\ Y &\sim \mathcal{N}(\mu_2, \sigma_2^2) \\ X, Y &\text{ independent} \end{aligned}$$



$$X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

(proof left to [Wikipedia](#))

$$\begin{aligned} \mathbb{E}[X+Y] &= \mathbb{E}[X] + \mathbb{E}[Y] \\ &= \mu_1 + \mu_2 \\ \text{Var}(X+Y) &= \text{Var}(X) + \text{Var}(Y) \\ &= \sigma_1^2 + \sigma_2^2 \end{aligned}$$

Holds in general case:

$$\begin{aligned} X_i &\sim \mathcal{N}(\mu_i, \sigma_i^2) \\ X_i &\text{ independent for } i = 1, \dots, n \end{aligned}$$



$$\sum_{i=1}^n X_i \sim \mathcal{N}\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$$

Back for another playoff game



What is the probability that the Warriors win?
How do you model zero-sum games?

$$P(A_W > A_B)$$

This is a probability of an event involving **two** random variables!

We will compute:

$$P(\underbrace{A_W - A_B}_{\text{A sum of Normals!}} > 0)$$

A sum of Normals!

Motivating idea: Zero sum games



Want: $P(\text{Warriors win}) = P(A_W - A_B > 0)$

Assume A_W, A_B are independent.

Let $D = A_W - A_B$.

$$= A_W + (-A_B)$$

$A_B \sim \mathcal{N}(-1470, (-1)^2 200^2)$

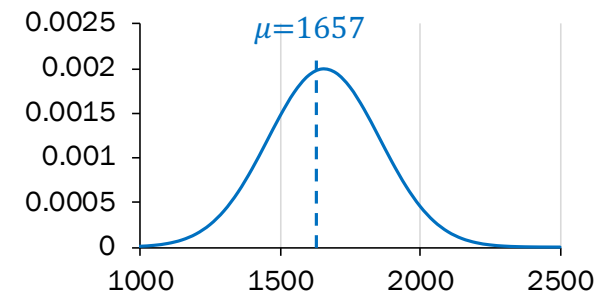
$\Downarrow$

$\mathcal{N}(-1470, 200^2)$

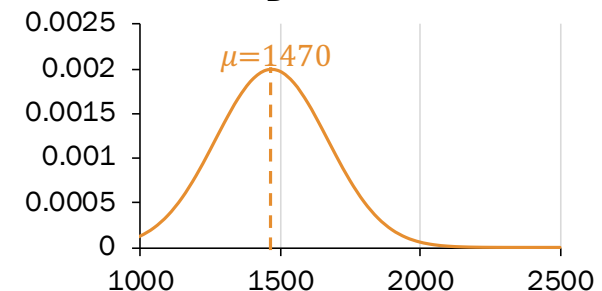
What is the distribution of D ?

- A. $D \sim \mathcal{N}(1657 - 1470, 200^2 - 200^2)$
- B. $D \sim \mathcal{N}(1657 - 1470, 200^2 + 200^2)$
- C. $D \sim \mathcal{N}(1657 + 1470, 200^2 + 200^2)$
- D. $D \sim \mathcal{N}(1657 + 1470, 200^2)$

Warriors $A_W \sim \mathcal{N}(S = 1657, 200^2)$



Opponents $A_B \sim \mathcal{N}(S = 1470, 200^2)$



Motivating idea: Zero sum games



Want: $P(\text{Warriors win}) = P(A_W - A_B > 0)$

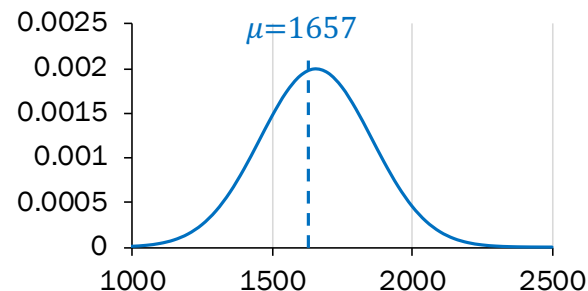
Assume A_W, A_B are independent.

Let $D = A_W - A_B$.

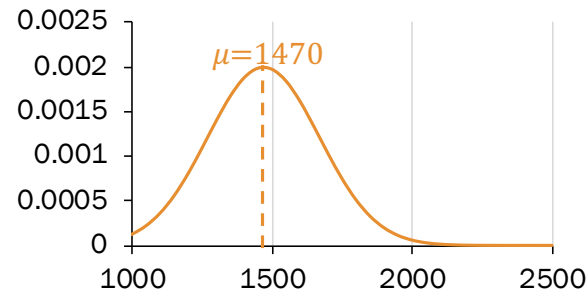
What is the distribution of D ?

- A. $D \sim \mathcal{N}(1657 - 1470, 200^2 - 200^2)$
- B. $D \sim \mathcal{N}(1657 - 1470, 200^2 + 200^2)$**
- C. $D \sim \mathcal{N}(1657 + 1470, 200^2 + 200^2)$
- D. $D \sim \mathcal{N}(1657 + 1470, 200^2)$

Warriors $A_W \sim \mathcal{N}(S = 1657, 200^2)$



Opponents $A_B \sim \mathcal{N}(S = 1470, 200^2)$



If $X \sim \mathcal{N}(\mu_1, \sigma^2)$,
then $(-X) \sim \mathcal{N}(-\mu, (-1)^2 \sigma^2 = \sigma^2)$.

Motivating idea: Zero sum games



Want: $P(\text{Warriors win}) = P(A_W - A_B > 0)$

Assume A_W, A_B are independent.

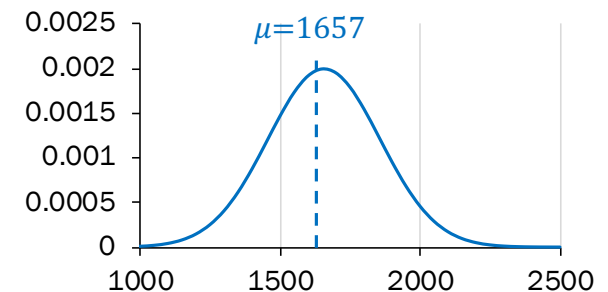
Let $D = A_W - A_B$.

$$\begin{aligned} D &\sim \mathcal{N}(1657 - 1470, 200^2 + 200^2) \\ &\sim \mathcal{N}(187, 2 \cdot 200^2) \quad \sigma \approx 282.842 \end{aligned}$$

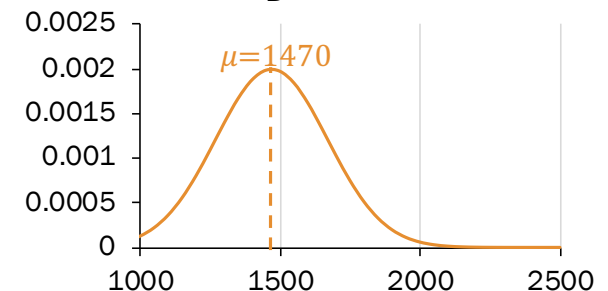
$$\begin{aligned} P(D > 0) &= 1 - F_D(0) = 1 - \Phi\left(\frac{0 - 187}{282.842}\right) \\ &\approx 0.74574 \end{aligned}$$

Compare with **0.7488**, calculated by sampling!

Warriors $A_W \sim \mathcal{N}(S = 1657, 200^2)$



Opponents $A_B \sim \mathcal{N}(S = 1470, 200^2)$



```
>>> from scipy.stats import norm
>>> 1 - norm(187, 80000 ** 0.5).cdf(0)
0.7457402843526317
>>> 1 - norm(0, 1).cdf(-187 / (80000 ** 0.5))
0.7457402843526317
```

Virus infections

Suppose you are working with the WHO to initiate a response to the onset of a virus. There are two exposed groups:

- G1: 20000 people, each independently infected with $p_1 = 0.1$
- G2: 10000 people, each independently infected with $p_2 = 0.4$

What is $P(\text{people infected} \geq 6100)$? An approximation is okay.

1. Define RVs & state goal

Let A = # infected in G1.

$$A \sim \text{Bin}(20000, 0.1)$$

B = # infected in G2.

$$B \sim \text{Bin}(10000, 0.4)$$

Want: $P(A + B \geq 6100)$

Strategy:

A. Sum of independent Binomials

B. Sum of independent Poissons

✓ C. Sum of independent Gaussians

D. Sum of independent Exponentials

no, $p_1 \neq p_2$



no, because $p_2 = 0.4$ is too large

} yes!!!

no, because mean and stdev of B are very different.

Virus infections

Suppose you are working with the WHO to initiate a response to the onset of a virus. There are two exposed groups:

- G1: 20000 people, each independently infected with $p_1 = 0.1$
- G2: 10000 people, each independently infected with $p_2 = 0.4$

What is $P(\text{people infected} \geq 6100)$? An approximation is okay.

1. Define RVs
& state goal

Let $A = \#$ infected in G1.

$A \sim \text{Bin}(20000, 0.1)$

$B = \#$ infected in G2.

$B \sim \text{Bin}(10000, 0.4)$

Want: $P(A + B \geq 6100)$

2. Approximate as sum of Gaussians

$A \approx X \sim \mathcal{N}(2000, 1800)$ $B \approx Y \sim \mathcal{N}(4000, 2400)$

$P(A + B \geq 6100) \approx P(X + Y \geq 6099.5)$ continuity correction

3. Solve

Virus infections

```
>>> 1 - norm(6000, 4200 ** 0.5).cdf(6099.5)
0.06235282662988528
>>> 1 - norm(0, 1).cdf((6099.5 - 6000)/(4200 ** 0.5))
0.06235282662988528
```

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Let $W = X + Y \sim \mathcal{N}(6000, 4200)$

$$P(W \geq 6099.5) = 1 - \Phi\left(\frac{6099.5 - 6000}{\sqrt{4200}}\right)$$

$$\approx 1 - \Phi(1.53531) \approx 0.06235$$

Sum of Independent Gaussians

$$\begin{array}{l} X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), \\ X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2) \\ X_1, X_2 \text{ independent} \end{array} \Rightarrow X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Is this related to linear transformations of Gaussians?

Recall:

$$\text{If } Y = aX + b, \text{ then } Y \sim \mathcal{N}(a\mu_X + b, a^2\sigma_X^2)$$

Teaser: Linear transforms vs. independence



Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = X + X$. What is the distribution of Y ?

- Are both approaches valid?

Independent RVs approach

Let $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$
be independent.

Then $Y = X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

Linear transform approach

Let $X \sim \mathcal{N}(\mu, \sigma^2)$.

If $Y = aX + b$,
then $Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.



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umm, X is NOT independent of X !

$$Y = X + X$$

$$X + X \sim \mathcal{N}(\mu + \mu, \sigma^2 + \sigma^2)?$$

$$Y \sim \mathcal{N}(2\mu, 2\sigma^2)?$$

Linear transform approach 

Let $X \sim \mathcal{N}(\mu, \sigma^2)$.

If $Y = aX + b$,
then $Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.

$$Y = 2X$$

$$Y \sim \mathcal{N}(2\mu, 4\sigma^2)$$

For independent $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$,
 $aX_1 + bX_2 + c \sim \mathcal{N}(a\mu_1 + b\mu_2 + c, a^2\sigma_1^2 + b^2\sigma_2^2)$